Pizza Sales Analysis

SQL Project using MySQL Workbench

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About the Project

- Dataset: Pizza Sales Data (21,350 orders)
- Tool Used: MySQL Workbench
- Database: pizzahut
- Tables Used: orders, orders_details, pizzas, pizza_types
- Goal: Analyze pizza sales data to answer key business questions using SQL queries

Database Structure

- **orders** → order_id, order_date, order_time
- **orders_details** → order_details_id, order_id, pizza_id, quantity
- **pizzas** → pizza_id, pizza_type_id, size, price
- **pizza_types** → pizza_type_id, name, category, ingredients

Tables are connected using Primary Key & Foreign Key relationships

Questions Solved

- **Basic**
  - Q1. Total number of orders placed
  - Q2. Total revenue generated
  - Q3. Highest priced pizza
  - Q4. Most common pizza size ordered
  - Q5. Top 5 most ordered pizza types

• Advanced

- Q6. Total quantity ordered by pizza category
- Q7. Distribution of orders by hour
- Q8. Category wise pizza distribution
- Q9. Average pizzas ordered per day

• Intermediate

- Q10. Top 3 pizzas by revenue per category
- Q11. Percentage contribution of each category to revenue
- Q12. Cumulative revenue over time

Retrieve the total number of orders placed.

```sql
select count(order_id) as total_orders
 from orders;
```

Result Grid

total_orders
21350

Calculate the total revenue generated from pizza sales.

```
select
round(sum(orders_details.quantity * pizzas.price),2) as total_sales
from orders_details join pizzas
on pizzas.pizza_id = orders_details.pizza_id
```

Result Grid

total_sales
817860.05

Identify the highest-priced pizza.

```sql
select pizza_types.name, pizzas.price
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
order by pizzas.price desc limit 1;
```

Result Grid | Filter Rows:

name	price
The Greek Pizza	35.95

Identify the most common pizza size ordered.

```
select pizzas.size, sum(orders_details.quantity) as total_quantity
from pizzas join orders_details
on pizzas.pizza_id = orders_details.pizza_id
group by size
order by total_quantity desc limit 1;
```

Result Grid | Filter Rows:

size	total_quantity
L	18956

List the top 5 most ordered pizza types along with their quantities.

```
select pizza_types.name, sum(orders_details.quantity) as total_quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id =pizzas.pizza_id
group by pizza_types.name
order by total_quantity desc limit 5;
```

Result Grid | Filter Rows:

name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Join the necessary tables to find the total quantity of each pizza category ordered.

```
select pizza_types.category, sum(orders_details.quantity) as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category
order by quantity desc;
```

Result Grid | Filter Rows:

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Determine the distribution of orders by hour of the day.

```
select hour(order_time) as hour,
 count(order_id) as order_count from orders
group by hour(order_time);
```

Result Grid | Filter

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198

Join relevant tables to find the category-wise distribution of pizzas.

```
select category, count(name) as distribution
from pizza_types
group by category;
```

Result Grid | Filter R

category	distribution
Chicken	6
Classic	8
Supreme	9
Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
select round(avg(quantity),0) as average_pizza from
(select orders.order_date , sum(orders_details.quantity) as quantity
from orders join orders_details
on orders.order_id = orders_details.order_id
group by orders.order_date) as order_quantity;
```

Result Grid

average_pizza
138

Determine the top 3 most ordered pizza types based on revenue.

```
select pizza_types.name,
sum(orders_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by revenue desc limit 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Calculate the percentage contribution of each pizza type to total revenue.

```
select pizza_types.category,
round(sum(orders_details.quantity * pizzas.price) / (select
sum(orders_details.quantity * pizzas.price)
from orders_details join pizzas
on pizzas.pizza_id = orders_details.pizza_id) * 100, 2) as revenue
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category
order by revenue desc;
```

Result Grid | Filter R

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Analyze the cumulative revenue generated over time.

```
select order_date,
round(sum(revenue) over (order by order_date), 2) as cumulative_revenue
from
(select orders.order_date,
sum(orders_details.quantity * pizzas.price) as revenue
from orders_details join pizzas
on orders_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = orders_details.order_id
group by orders.order_date) as sales;
```

Result Grid | Filter Rows:

order_date	cumulative_revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select category, name, revenue
from
(select pizza_types.category, pizza_types.name,
sum(orders_details.quantity * pizzas.price) as revenue,
rank() over (partition by pizza_types.category order by
sum(orders_details.quantity * pizzas.price) desc) as rn
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a
where rn <= 3;
```

Result Grid | Filter Rows: | Export:

category	name	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.70000000065
Veggie	The Mexicana Pizza	26780.75
Veggie	The Five Cheese Pizza	26066.5

Thank You